Lyapunov-based MRAC of an Unstable Plant

Adaptive Control, Homework 6

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English summary of my original Persian report, which is kept as a PDF in this folder.

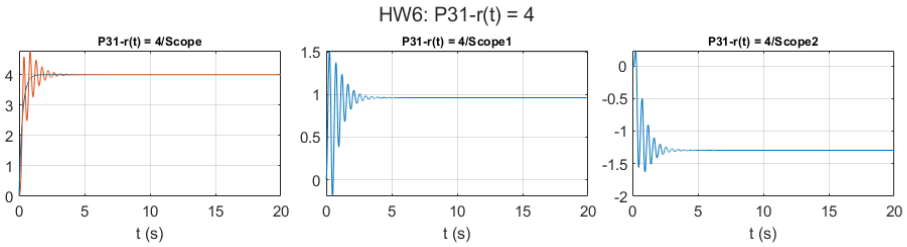
All cases are subsystems of `HW6.slx`. They reproduce two examples from the lecture slides.

Slide 31

The plant is $\dot{x} = x + 3u$, which is unstable and whose parameters are unknown to the controller. The reference model is $\dot{x}_m = -4x_m + 4r$. The controller is $u = \hat{k}_x x + \hat{k}_r r$, and both gains adapt with Lyapunov-based laws driven by the tracking error. I simulated a constant reference $r = 4$ and a sinusoidal one, and plotted the output against the reference model together with $\hat{k}_x$ and $\hat{k}_r$.

Slide 34

The same structure with an extra adapted parameter $\hat{\theta}$, a matched uncertainty. I plotted the output and the three estimates $\hat{k}_x$, $\hat{k}_r$ and $\hat{\theta}$.



Re-run in R2023b: slide 31 with $r = 4$, output against the reference model, $\hat{k}_x$ and $\hat{k}_r$.